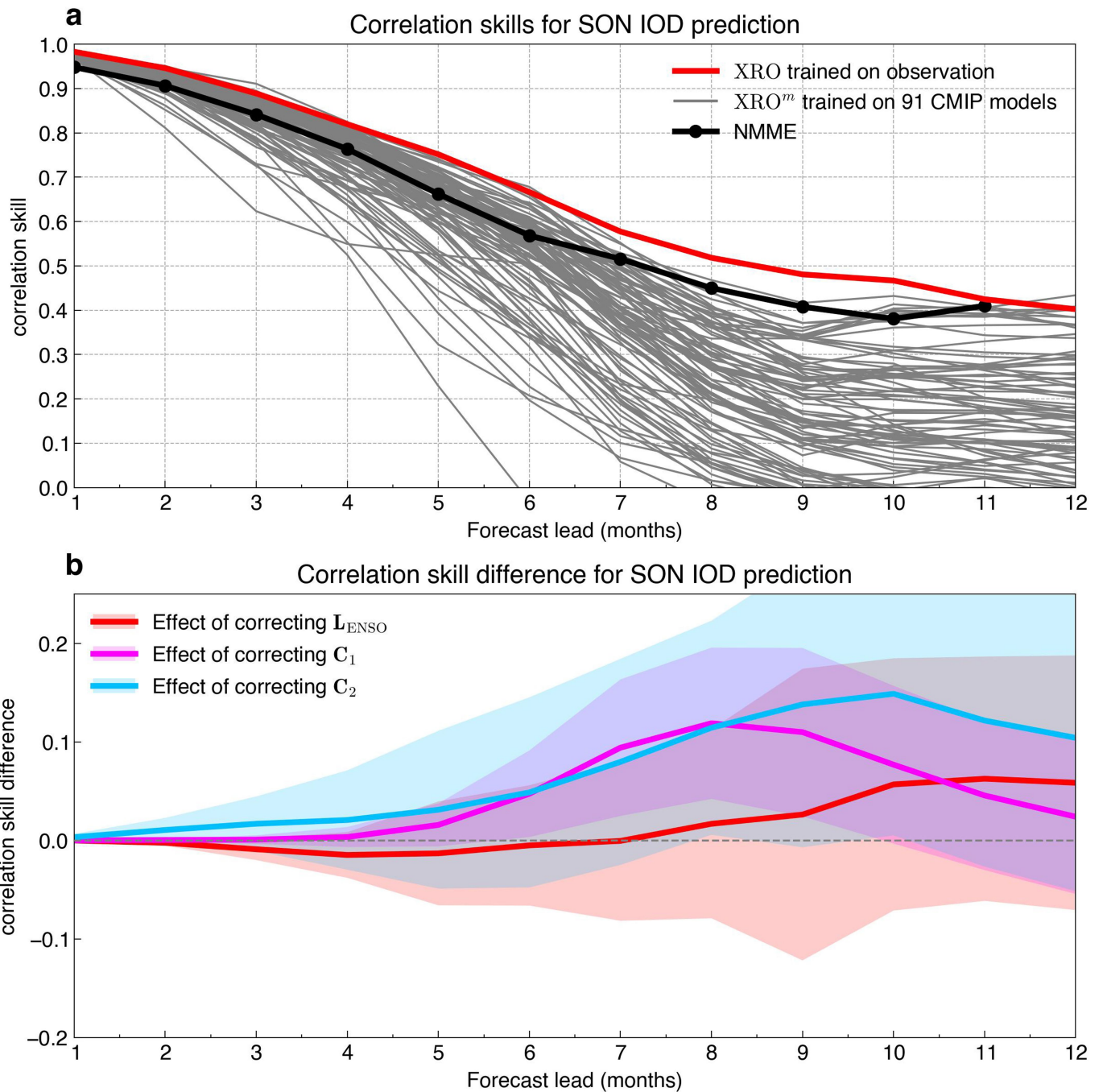




Extended Data Fig. 8 | Correlation forecast skill for the Indian Ocean Dipole, using the XRO trained with climate model outputs. (a) The correlation skill of the IOD index in Sep-Oct-Nov (SON) as a function of forecast lead, in the XRO^m trained solely on 91 individual CMIP model outputs (grey curves), the XRO trained on observations (red curve), and the original (not XRO) multi-model mean of the ensemble means of the forecasts from the NMME models (black). (b) the ensemble mean and 10%–90% spread band of the

changes in correlation skill for the IOD index, obtained by correcting the ENSO internal linear dynamics ($XRO_{L_{\text{ENSO}}}^m - XRO^m$, red), or the remote-mode feedbacks onto ENSO ($XRO_{C_1}^m - XRO^m$, magenta), or the ENSO teleconnections to remote modes ($XRO_{C_2}^m - XRO^m$, blue). Reforecasts using the XRO trained on climate model output show that reducing CGCM biases in the dynamics of ENSO's climate mode interactions improves IOD forecasts.